
Artificial Intelligence

BS (CS) Spring 2026

Lab 08 Tasks



Learning Objectives:

1. Constraint Satisfaction Problem

Lab Tasks

Submission Instructions

1. Create one notebook file for all questions with name *ROLLNO_SEC_LAB8_II* e.g. **i23XXXX_A_LAB08**
2. Submit your .ipynb file with correct name on Google Classroom under your section heading.
3. If you don't follow the above-mentioned submission instruction, you will be marked **zero**.

Plagiarism in the Lab Task will result in **zero** marks in the whole category.

*Note: **You must implement the core logic of the solutions by yourself.** The Python Constraint Library is just for reference since it abstracts away the core logic.*

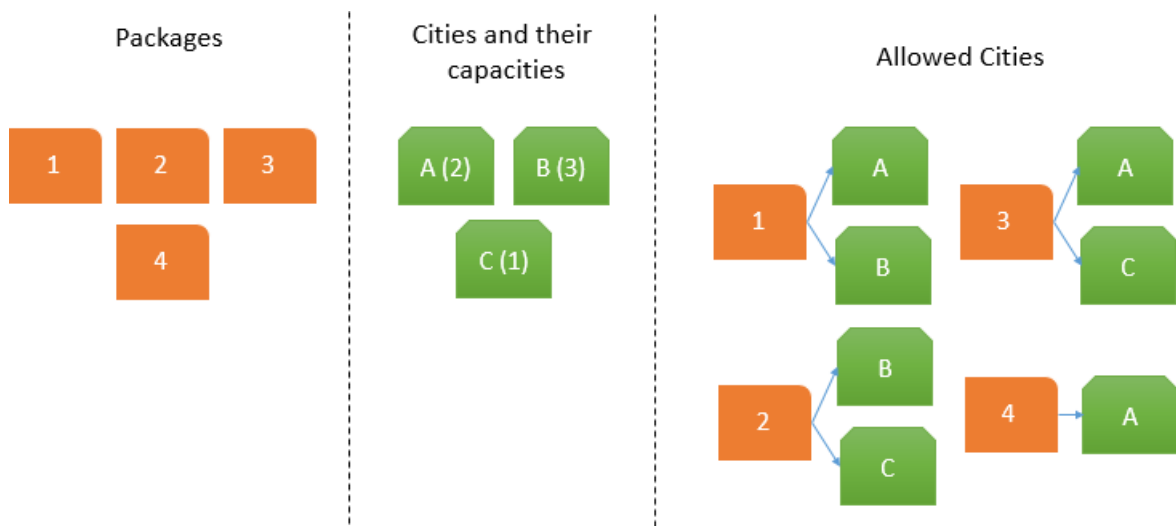
Question

Suppose you are a delivery manager at a logistics company that needs to distribute packages to different cities in the country efficiently. You have a list of packages and cities, and you need to assign each package to a city such that each package is delivered to a city only once and each city can receive multiple packages.

The inputs to the program will be a list of packages, and a list of cities, each with its capacity to receive packages. The program should output the optimal assignment of packages to cities that satisfy all the constraints.

Suppose we have the following input data:

1. packages = [1, 2, 3, 4] (i.e., four packages to be delivered)
2. cities = ['A', 'B', 'C'] (i.e., three cities to deliver packages to)
3. city_capacity = 'A': 2, 'B': 3, 'C': 1 (i.e., each city can receive a certain number of packages)
4. allowed_cities = 1: ['A', 'B'], 2: ['B', 'C'], 3: ['A', 'C'], 4: ['A'] (i.e., some packages can only be delivered to specific cities)



Variables and Domain

We can use a decision variable $x_{i,j}$ to denote whether package p_i is assigned to city c_j or not. The domain for each variable $x_{i,j}$ is 0,1, where 0 represents that package p_i is not assigned to city c_j , and 1 represents that package p_i is assigned to city c_j .

Constraints

1. Each package should be assigned to exactly one city
2. Each city should receive packages within its capacity

Your goal is to develop a Python program that can solve this constraint satisfaction problem by finding a feasible solution that satisfies all the constraints and minimizes the overall cost of delivering the packages.

Answer:

{'A': [4, 1], 'B': [2], 'C': [3]}